Balaji Vijayan

balajivijayan161@gmail.com

PROFESSIONAL SUMMARY

Aspiring ECE undergraduate student with a strong foundation in electronics, programming, and problem-solving. Skilled in C++,Python, Java and SQL, with hands-on experience in academic projects and teamwork.

IMPLANT EXPERIENCE

Megawin Switchgear Private Limited - **Switchgear Engineering Intern**
Salem, India
01/2026 - 02/2026

- Worked with the engineering team on the design, assembly, and testing of medium-voltage switchgear systems.
- Acquired hands-on experience with vacuum circuit breakers (VCBs) and load break switches during practical training sessions.

Tech Forge - **Full Stack Intern**
Salem, India
05/2025 - 06/2025

- Conducted code reviews, testing, and debugging processes.
- Utilized Git/GitHub for version control and project collaboration to streamline development workflow.

SAIL SALEM - **Central Electrical Maintenance Intern**
Salem, India
12/2024 - 01/2025

- The training focused on industrial safety protocols, covering workplace safety, hazard identification, and emergency response to meet regulatory standards.

PROJECTS

Inhalo Tracker – Breath-Based Health Monitoring System

- Breath analysis system using ESP32 & MQ-series sensors to detect biomarkers (acetone, ammonia, H_2S, hydrogen) with ADC signal processing, real-time display & IoT remote monitoring for early health screening.

Hybrid Footsteps Energy Harvesting Tile Using Piezoelectric And Triboelectric Mechanism

- Hybrid footstep energy harvesting tile combining piezoelectric & triboelectric mechanisms with rectification, supercapacitor storage & DC-DC boost converter to power low-power loads in smart public spaces.

TECHNICAL SKILLS

- SQL,C++,Python and Java
- Working knowledge in IoT applications and embedded systems
- Networking Technologies

EDUCATION

Expected in 05/2027
Sri Krishna College Of Engineering And Technology
Coimbatore

Bachelor's Degree: Electronics And Communication Engineering
CGPA - 8.15 (Full Time)

04/2023
Senthil Public School
Salem

HSC
Percentage : 74.2 (Full Time)

04/2021
Notre Dame of Holy Cross
Salem

SSLC
Percentage : 90 (Full Time)

CERTIFICATIONS

CISCO - CCNA Introduction to Networks

INFOYSYS - Java, C++ and SQL

CO-CORRICULAR PARTICIPATION

- Coordinated IEEE UPSKILL Event - 2024
- Participated in IEEE WIE International Leadership Summit - 2024
- Organized KRISNECS - 2024
- Organized and Participated in IEEE XTREME Coding Contest - 2025
- Member of Organizing Panel - IEEE ICINVENTS - 2025